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Shaping AI-driven language learners' learning approaches, and motivation through computational thinking skills in computer-assisted language learning (CTCALL): A bisymmetric research design

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Abstract

With the development and integration of information and communication technologies, the learning needs and skills have revised and refined, and 21st-century digital skills have recently become more important than before. Computational thinking skill, the heart of the 21st-century digital skills, has recently been developed and validated in the field of computer-assisted language learning (CTCALL), but has received little attention in CALL. Consequently, the researcher cultivated this important skill in Intelligent CALL (ICALL) for 104 Iranian EFL learners and explored it on the language learners' motivation and their learning approaches to ICALL. The bisymmetric research design validated the factorial validity of this skill and showed that the abstraction and evaluation skills were among the necessary conditions to shape language learners' deep approach to ICALL, and further level of them can shape their deep approach to it as well. However, generalization was not among the necessary conditions, but the higher language learners can identify patterns in solving specific language tasks with AI, and applying those patterns to other language tasks involves new ways of thinking and cross-referencing ideas with AI, culminating in maximizing their ICALL for learning language. Based on these findings, the study provides a new conceptual model to the field of CALL and recommends that teachers adopt a problem-solving approach within ICALL and cultivate new skills, such as CTCALL, in their language learning environment to foster motivation and deep, organized approaches to ICALL.

Keywords: Artificial Intelligence in Education, Students approaches to learning environments (SAL) 21st-century digital skills, L2 motivational self-system, intelligent CALL (ICALL)

Introduction

Technological advancements and the advent of Artificial Intelligence (AI) have resulted in numerous changes across many sectors, particularly in education and CALL. In effect, the mode of language teaching and learning has shifted from being

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centred on the teacher or student to personalized language learning with AI, and the process of teaching with technology has changed to a problem-solving one (Rahimi & Sevilla-Pavón, 2025). The availability of AI and Chatbots in the modern world allows students to use them to solve their personal and academic problems. These developments have prompted several pedagogical organizations, including the Assessment and Teaching of Twenty-First Century Skills(ATC21S), the Organization for Economic Cooperation and Development (OECD), the European Union (EU), the Computer Science Teachers Association (CSTA), the International Society for Technology in Education (ISTE), and the International Association for the Evaluation of Educational Achievement (IEA), to revise curricula and learning objectives where they recommended, and stressed the importance of acquiring learners as well as future generations 21st-century digital skills to ensure confidence, critical thinking, and responsible engagement with digital technologies as well as AI in their daily lives and future occupations (Agbo et al., 2023). These skills include Computational Thinking (CT), Design Thinking (DT), algorithmic thinking, information literacy, digital problem-solving, and higher-order thinking (Rahimi & Sevilla-Pavón, 2025).

In the current age of innovation, 21st-century digital skills such as Computational Thinking skills and Design Thinking skills which have a problem-solving approach, are becoming increasingly popular since learners can achieve reliable results through the use of them as well as solve the challenges posed by AI inaccuracies and hallucinations (Rahimi & Sevilla-Pavón, 2025b), which has become the main goal of stakeholders to address this issue in Artificial Intelligence in Education (Crompton & Burke, 2024). These important skills have not been given much focus by CALL researchers, who recently suggested that researchers in CALL and applied linguistics should take a wider view on CALL materials and the use of AI in language education, going beyond just teaching language skills to also include 21st-century digital skills in their language classes (Rahimi, 2023; Rahimi & Sevilla-Pavón, 2025a).

Computational thinking skill, known as the heart of 21st-century digital skills, refers to 'formulating problems so their solutions can be represented as computational steps and algorithms' (Aho, 2012, p.832). It is a universally applicable skill set and attitude for everyone (Wing, 2006, p. 33), and hence, there is a growing demand to integrate it into compulsory education systems worldwide to equip students with the skills needed to thrive in the digital age. For this sake, several fields, particularly Science, Technology, Engineering, and Mathematics (STEM) integrated it into their curriculum, and their researchers have declared that this critical skill is specialized to their field (Chevalier et al., 2020); however, recently Rahimi and Sevilla-Pavón, (2025a) developed the theory of Computational Thinking Skills in Computer-Assisted Language Learning (CTsCALL), with its corresponding scale to the field of CALL and applied linguistics, and validated that this crucial 21st-century digital skill can apply to the humanistic field like CALL and Applied Linguistics. However, this emerging theory and skill must be integrated and validated across various disciplines within CALL to assess its predictive capacity in enhancing language learners' performance. Prominent figures in the CALL field have also emphasized that as technology continues to evolve, the discipline should keep advancing accordingly (Colpaert, 2020; GimenoSanz, 2015).

There is currently a limited understanding of the role of CT in shaping learners' psychological disposition within education more broadly, as well as in the context of language acquisition (Kong & Wang, 2023; Rahimi & Sevilla-Pavón, 2025 a). Furthermore, a recent systematic literature review on CT indicates that the majority of studies assessing learners' CT have predominantly focused on programming and computing skills, rather than exploring its implications in other educational areas (Tang et al., 2020; Weng et al., 2024). Accordingly, the above recommendations, gaps, and developments in new skill sets in our field led the researchers to integrate the CTCALL in Intelligent CALL (ICALL) and explore its effects on language learners' performance in this field. Even so, it seems that a certain psychological disposition is necessary to cultivate any skill in second language learning. Along with the advances in technology and artificial intelligence, language learners' psychological aspects have also evolved, which has led to the development of new terms such as L2 grits (Teimouri et al., 2020), Artificial Intelligence L2 motivational self-system (Rahimi & Sevilla-Pavón, 2025), or digital self-authenticity (Smith et al., 2020). A new psychological concept recently applied and validated to the field of CALL, a concept called Students Approaches to Learning Environment (SAL), was developed by Marton and Säljö (1976) and has been recently incorporated and validated into the field of CALL and Language Massive Open Online Courses (LMOOCs) by Rahimi (2024). According to this theory, learners have three approaches, namely surface, deep, and organizing, to any learning environment. Due to its pedagogical orientation in exploring language learners' learning behaviours in any learning environments, the researchers chose this theoretical framework for the current study because recent CALL studies have examined learners' attitudes, perceptions, and experiences about ICALL, especially through the lens of the technology acceptance models (An et al., 2023; Assassi, 2025; Creely et al., 2025; Zou et al., 2023). Since these studies focus on subjective norms, ease of use, and the perceived usefulness of AI, factors that are emotionally and technologically driven, a pedagogically oriented framework, such as SAL to ICALL, is necessary to offer deeper pedagogical and practical insights into the field of CALL.

Recent systematic and integrative literature reviews have also emphasized this need by reporting that studies have explored learners' experiences, perspectives, emotions, and attitudes toward ICALL and suggesting that more trends are necessary in this area (Kartal & Yeşilyurt, 2024; Koç & Savaş, 2024; Meina & Chaoran, 2025). To heed these recommendations and shift the focus from the emotional aspects of language learners in AILL to their pedagogical dimensions, the researchers will apply the SAL framework in this study as well. Motivation among language learners is a crucial psycholinguistic factor that researchers often identify as both a prerequisite and a mediator, enhancing language learning performance and influencing other psycholinguistic traits in various SLA contexts. However, its effects can vary depending on the language learning contexts and its relationship with learning skills and psychological elements. According to the theory of complex dynamic systems, learners' psychological traits, like motivation and self-regulation, are interrelated and mutually dependent in every language learning environment (Mercer, 2018). Additionally, these relationships have been examined concerning learners' skills and subskills (Paradowski & Jelińska, 2023). However, the lack of research on how these psychological factors connect with new digital skills like CTCALL and other psychological traits like learners' approaches to ICALL shows a gap that this study might

fill. Consequently, we will test a model that shows the complex correlations between language learners' CTCALL skills and their AI L2 motivational self-system in the context of ICALL.

Literature review

Computational thinking skills in computer-assisted language learning (CTsCALL)

Building on Papert's pioneering work from the last century (Papert, 1980), many researchers have considered the effects of CT on education. Jeannette Wing has become a key figure in CT, as she was the first to connect it with problem-solving skills that are relevant not just to computer scientists but to all individuals (Wing, 2006). Wing defined CT as "the thought processes involved in formulating problems and their solutions so that the solutions are represented in a form that can be effectively carried out by an information processing agent" (Wing, 2011). In the past two decades, numerous researchers have faced the challenge of defining this skill, with its subskills, given the nature of the CT and study context. Typical examples of computational thinking skills comprise abstraction, algorithmic thinking, cooperation, creativity, critical thinking, data analysis, debugging, decomposition, heuristic reasoning, and problem solving (Agbo et al., 2023; Doleck et al., 2017; Jha & Tsai, 2026; Yusuf & Noor, 2024). There is also a challenge for a subset of researchers in distinguishing between computational thinking skills, problem-solving skills, and critical thinking skills. In general, computational thinking skills refer to a set of cognitive abilities related to computational tasks and activities, especially those involving information and communication technology. Unlike critical thinking and problem-solving, CT focuses on breaking down problems and formulating solutions that can be executed by humans or machines by using algorithms, abstractions, and decomposition (Agbo et al., 2023; Doleck et al., 2017; Rahimi & Sevilla-Pavón, 2025a), and it explores the process of creating a systematic, step-by-step solution using computers. Conversely, critical thinking is characterized by analyzing, evaluating, and judging to answer the questions "why" and "how" (Doleck et al., 2017). Moreover, problem solving is "finding a way out of a difficulty, a way around an obstacle, attaining an aim that was not immediately understandable" (Polya 1981). As cited in Doleck et al., (2017, p. 6). Nonetheless, some researchers have posited that critical thinking skills and problem-solving skills are subskills of CT (Doleck et al., 2017; Korkmaz et al., 2017).

In view of the different theoretical frameworks and definitions of CT, we followed Rahimi and Sevilla-Pavón (2025a), who developed the CTCALL theory and its corresponding scale for the field of CALL and applied linguistics to enhance the validity and predictive power of our field. Hence, we operationally define the following CTCALL components in ICALL:

- *Algorithmic Thinking (AL)* Language learners' skills to solve language tasks step-by-step using AI.
- *Abstraction (AB)* Language learners' skills to employ deductive reasoning by focusing on key information rather than details while solving language tasks with AI.
- *Collaboration (CL)* Language learners' skills in collaborating with others to solve language tasks together using AI.

- *Evaluation (EV)* Language learners' skills to compare various solutions and responses provided by AI and identify the most effective option.
- *Generalization (GE)* Language learners' skills to identify patterns in solving specific language tasks with AI and apply those patterns to other language tasks and contexts in the target language.

Artificial intelligence L2 motivational self-system

A prominent area of L2 studies has consistently been motivation, especially Dörnyei's (2005) L2 motivational self-system. Recently, globalization has increased the popularity of the concept of English as a lingua franca (ELF) in the EFL context. Culminating in groundbreaking findings by Dörnyei (2009), it was claimed that actual language use, rather than integrativeness, shapes students' motivation in EFL contexts because they lack direct interaction with the target language community. Thus, the L2MSS theory shifts away from the traditional socio-psychological model of language learning motivation by focusing on three key components: the ideal L2 self, the ought-to L2 self, and the L2 learning experience. The integration of ICT has further developed this theory, enabling L2 scholars to refine and broaden its constructs to align with the evolving L2 self-identities of learners, as noted in prior studies. To clarify, Henry (2013) observed the authenticity gap by reporting that learners demonstrated strong motivation for digital language learning, while their motivation and effort in traditional face-to-face classroom settings diminished. This gap between traditional language learning and digital learning is addressed by Henry and Cliffordson (2017) in a Swedish context, resulting in the addition of a new construct to the L2MSS theory, namely an authenticity gap. To continue their path, Smith et al. (2020) validated their findings in the Chinese EFL context in online language learning; however, they provided a new construct as digital self-authenticity.

Similarly, Thorsen et al. (2020) found a discrepancy between the learners' future self and the current self, as L2MSS theory did not adequately address language learners' present L2 self. This prompted them to incorporate the actual and current L2 self into the theoretical framework. However, additional research is necessary to explore these new constructs across different language contexts and various skills and learning behaviors to evaluate their predictive power in measuring overall motivation among language learners. Consequently, we operationalized the following constructs as follows:

Ideal L2 self (IDL)

Language learners' personal and individual motivation to use AI for language schooling to reach their future objectives.

Ought-to L2 self (OUT)

Learners' attributions to learn language with AI to fulfill expectations or avoid negative consequences, like failing their English course.

Current L2 self (CU)

The self-evaluations and assessments made by language learners regarding their actual language skills and competence to learn with AI.

Digital self-authenticity (DSA)

A comparison of language learners' motivation while using AI for language schooling with their past experiences in different learning settings, including traditional and online formats.

Language learners' approaches to ICALL

Research on SAL has a long history, beginning with Marton and Säljö (1976), who identified two levels of information processing: deep and surface processing. This concept evolved into "approaches to learning," and by the time it was categorized into three approaches (Entwistle, 2017), the surface approach was recognized as a strategy where learners memorize and reproduce material primarily to pass a course. This approach has also been linked to students' difficulty in recognizing connections between ideas or concepts.

The deep approach, on the other hand, involves students aiming for a comprehensive understanding of the subject by applying critical thinking skills and engaging in meaningful learning to form a personal grasp of the material. The third approach, known as the organizing approach (formerly the strategic approach), pertains to students' daily study habits, focusing on how they manage their time and structure their studies.

With the advancement of technology, SAL has been validated, revised, and extended through previous studies in blended learning (Ellis & Bliuc, 2015), online learning (Han & Geng, 2023), and language MOOCs (LMOOC: Rahimi, 2024). However, in comparison with other psycholinguistic factors, like self-regulation and motivation, SAL has been overlooked by CALL scholars, which has been recommended by Rahimi (2024) to evaluate its predictive validity in CALL. Moreover, it explores technology user intention to employ the target technology in a more pedagogical-psychological manner than emotional-psychological ones, such as TAM.

From a theoretical perspective, instead of merely specifying whether learners intend to use a specific technology, SAL provides a cognitive-pedagogical lens explaining how learners process and regulate learning. Contrary to technology acceptance frameworks highlighting perceptions such as usefulness or ease of use, SAL emphasizes the depth of cognitive engagement, metacognitive regulation, and strategic effort that learners invest in task completion (Biggs, 1987; Entwistle, 2017). In ICALL, as an AI-supported environment, learners are constantly required to deal with interpreting AI feedback, evaluating multiple outputs, identifying inaccuracies, and making informed decisions, all of which closely align with computational and higher-order thinking processes. Therefore, it is not surprising to consider the acquisition of CTCALL skills to be of high significance as it shapes learners' information processing patterns, which in turn foster deeper or more organized learning approaches. Grafting digital problem-solving skills and motivational self-perceptions to observable learning behaviors and outcomes, SAL functions as a mediating mechanism. Accordingly, incorporating SAL within ICALL research provides a theoretically robust framework for explaining how cognitive competencies and

motivational orientations are transformed into qualitatively distinct approaches to AI-mediated language learning, thereby moving beyond technology acceptance accounts toward pedagogically substantive learning processes (see Biggs & Tang, 2011). These considerations guided the development of the operational definitions for the following factors:

Surface Approach Language learners limit their use of AI, viewing it as promoting passive cognitive processing that focuses on minimal and rote learning.

Deep Approach Effective use of AI fosters new ways of thinking and enables learners to cross-reference ideas, resulting in greater engagement.

Organizing Approach Language learners integrate AI into their daily academic routines as a valuable tool for language learning.

Literature review and its gap

A large body of research has embraced and integrated CT to acquire innovative and complex problem-solving skills, primarily through AI. Several studies indicated that learners who acquire CT can apply their ideas, creativity, and imagination to practical situations, particularly through LLMs and AI (Yu et al., 2024). A study on virtual exchange by Rahimi and Sevilla-Pavón (2025a) addresses this issue in language education and CALL, in which Spanish language learners solve their language task through CTCALL. The researchers reported that learners who applied problem-solving skills, such as abstraction, where they focused on obtaining key information rather than on details, and identified patterns to solve their language tasks, perceived CTCALL as appropriate for their skills, language course, and VE environment. Another study conducted by Parsazadeh et al. (2020) cultivated CT in language learners through the use of digital storytelling. The researchers discovered that this skill not only improved language learners' performance but also boosted their intrinsic and extrinsic motivation. Additionally, scholars have noted that CT skills, such as decomposition and abstraction, positively influence foreign language acquisition, particularly in areas like grammar and writing (Yu et al., 2024). However, the impact of these problem-solving skills in ICALL, where learners tackle language tasks with AI, potentially encountering inaccuracies and bugs, on their learning behaviour and performance has yet to be explored.

Moreover, CALL studies have indicated that learners' motivation in general, and L2 self-identities in particular, are among the prerequisite factors in digital language learning contexts for shaping language learners' learning performance and psychological well-being. For clarification, Smith et al. (2020) found that Chinese EFL learners' ideal and current L2 selves notably increased their efforts in online language learning. Similarly, Rahimi (2023) reported that both ideal and ought-to L2 selves influenced Iranian EFL learners' intentions to study a language in LMOOCs. This effect also extended to shaping learners' deep and surface approaches in LMOOC, serving as a mediating variable between their perceived support and learning approaches to it.

However, in ICALL, these self-identities, except digital self-authenticity, did not mediate the relationship between Spanish language learners' DEAILL skills and their L2 grit (Rahimi & Sevilla-Pavón, 2025b). These findings may be attributed to the context-specific nature of motivational factors, which are influenced by contextual elements such as the learning environment and course structure (You & Dörnyei, 2016). Additionally,

a deficiency in opportunities for autonomy, personal expression, and new skills may also play a role (You & Dörnyei, 2016). This presents an opportunity for further investigation into the relationships between language learners' motivational factors, CTCALL, and their learning approaches in ICALL, where learners can exercise greater autonomy and creativity through the integration of CTCALL.

Due to the development and integration of AI, new skills such as CTCALL, creativity, autonomy, and the personalized nature of language learning are experiencing new trends, which require further exploration. Prior literature reviews have addressed these gaps, indicating a connection between critical thinking and language education, though additional research is required to further elucidate this relationship (Li et al., 2024; Rahimi & Sevilla-Pavón, 2025a; Yu et al., 2024). Moreover, the role of CTCALL in shaping language learners' intention to use VE (Rahimi & Sevilla-Pavón, 2025a) and their motivation in digital storytelling (Parsazadeh et al., 2020) has been explored, but further study is needed to explore CTCALL with other factors, like learners' L2 self-identities, and their approaches to another field, such as ICALL. By doing this, a new conceptual model and serial relationships can be added to the complex dynamic systems and CALL. For these reasons, the researcher will answer the following research questions:

RQ1: What are the factorial structures of CTCALL in ICALL and Iranian EFL contexts?

RQ2: To what extent can EFL learners' CTCALL skills predict their Artificial Intelligence L2 motivational self-system, and learning approaches in ICALL?

RQ3: To what extent can Artificial Intelligence L2 motivational self-system mediate the relationship between learners' CTCALL and their approaches to ICALL?

RQ4: Which one of the CTCALL and AIL2MSS variables will be among the necessary conditions to shape language learners' deep and organizing approaches to ICALL?

Methodology

Study procedure and its participants

The study collected data from 104 voluntary Iranian EFL students at a state university in Tehran, aged between 19 and 25, majoring in various fields, who had enrolled in two classes focused on reading comprehension skills. See Table 1 for the participants' demographic information.

Table 1 Participants' demographic information

		N	%
Gender	Male	0	0.00
	Female	104	100
Age	18–22	92	88.4
	23–26	12	11.6
	26 <	0	0
	Years		
Language	1–2	56	53.8
Learning	3–5	30	28.8
Experience	4 <	18	17.3

Using AI tools and video projectors, the second author facilitated classroom procedures to cultivate CTCALL. These tools helped guide students in their tasks, display videos, present AI-generated multidimensional texts, and analyze student responses to each activity. At the beginning of the course, during the first session, students took the Oxford Placement Test (OPT) to assess their average language proficiency, which informed the selection of suitable texts. The texts were sourced from the course textbook, *Focus on Vocabulary 1: Bridging Vocabulary* (Pearson Longman).

The CTsCALL skills were employed in all three phases of instruction: engagement, study, and personalization. To employ *algorithmic thinking*, students were guided step-by-step through tasks for all texts covered in class. For instance, to infer the general idea of a text, they first examined AI-generated vocabulary clouds based on a corpus compiled from the target text. The most frequent words, visually emphasized by a larger size, helped them make initial predictions. In other words, more frequent vocabulary, as the focus of the discussion in the text, made up a considerable portion of the text. The students could click on these words to view contextual sentences or concordances related to the word in the passage. Next, they refined their guesses using the text's title and accompanying image. Simultaneously, they learned to leverage the first and last sentences of paragraphs, noting transitional phrases (e.g., however, but, albeit, on the other hand) to identify key ideas. By combining these strategies, they employed *abstract thinking* rather than reading the entire text in detail. Finally, collaboratively, they analyzed the texts and presented joint responses. At the beginning of each chapter, these steps were followed in order. Other tasks for each text followed a sequential order, reinforcing the principle that progressing through the chapter depended on completing one task before advancing to the next.

As another illustration of *algorithmic thinking*, students employed a structured approach to determine the genre of a given text. Following a defined sequence, they first analyzed AI-generated vocabulary clouds, focusing on the most frequent words. Next, they examined sentences containing these high-frequency terms represented by larger words in the cloud. Afterwards, by integrating their prior knowledge of potential genres with contextual clues from the selected sentences, they collaboratively deliberated within their groups. Ultimately, they reached a consensus and presented their final response.

Several tasks focused on developing *evaluation* skills. For instance, to reinforce challenging vocabulary from the chapter, students completed AI-generated gap-fill exercises requiring them to select appropriate synonyms. They also used AI-assisted corpus tools to collaboratively examine vocabulary usage across different genres, analyze word collocations, and identify frequent synonyms through AI-generated word clouds. Working in groups, they evaluated possible responses, selected the most appropriate vocabulary, and justified their choices. Another task required students to paraphrase specific phrases or sentences containing target expressions. They collaborated to interpret meanings using AI tools like ChatGPT and DeepSeek, consulting more proficient group members when needed. Students cross-referenced online dictionaries and Google Translate to assess their interpretations before presenting their joint conclusions. Through these activities, they developed the ability to critically evaluate and synthesize information from multiple resources to arrive at optimal solutions.

Learners were indirectly led to apply *generalization* skills through various tasks requiring them to apply learned concepts to new contexts. Initially, individuals drew upon their background knowledge to share opinions about effective text summarization techniques. The instructor provided multiple instructional documents and practical videos demonstrating summarization stages with concrete examples. Building on these resources, students collaboratively practiced producing summaries for assigned texts, consciously applying previously acquired skills. This cyclical process was repeated for each text, allowing students to reference prior summarization experiences and incorporate feedback from earlier class discussions. During group work, students critically evaluated draft summaries before presenting their final versions to peers. They actively engaged with classmates' judgments and comments, fostering reflective learning.

The course also incorporated training on using AI tools effectively: students learned to utilize Grammarly's free version to review different categories of writing suggestions for their summaries. Additionally, they received instruction on crafting effective prompts for AI platforms to analyze their summary writing. Importantly, students were taught to critically assess AI-generated feedback rather than accepting all suggestions at face value. They developed discernment in deciding which recommendations to implement and which to disregard based on their understanding of effective summarization principles. For subsequent sessions, students were required to apply these skills independently by using AI tools to refine their summaries before submission on the designated platform. This structured approach progressively built their capacity to generalize summarization techniques while critically engaging with both peer and AI feedback.

Throughout the semester's chapters, students developed their text comprehension and task completion through interaction with diverse digital resources. These included AI-generated images and media, instructional videos, multimedia websites, online search engines, and AI-assisted corpus tools. This *collaborative discovery process* particularly enhanced the three key phases. The collaborative approach manifested differently at each stage. During engagement, while trying to make informed predictions, students pooled their knowledge to hypothesize about thematic content. In the practice phase, while connecting AI-generated media to reconstruct complete stories using chapter vocabulary and phrases, the students collectively synthesized multimedia elements with textual components to build coherent narratives. For personalization tasks, the groups analytically related the text content to personal experiences or identified cultural parallels/differences between the text's context and their own and worked together to draw meaningful connections between academic content and real-world applications, while also examining cross-cultural perspectives. The topics were carefully selected to align with both the students' language proficiency levels and cultural backgrounds, ensuring the material was accessible yet challenging. This multimodal approach, combining AI tools with collaborative learning strategies, enabled students to develop deeper textual understanding while acquiring practical digital literacy skills applicable across various learning contexts.

Notably, students initially applied abstraction skills to predict text genres and identify main ideas before employing three core competencies: algorithmic thinking through structured task completion steps; collaboration to implement ideas and create required products; and evaluation using AI tools and corpus-assisted resources to refine responses. This approach enabled systematic engagement with diverse reading

comprehension tasks. Students practiced essential strategies, including skimming, scanning, paraphrasing, and summarizing, while developing specific analytical skills like pronoun reference identification and content prediction for subsequent paragraphs, and employed generalization skills to similar tasks in each chapter. Throughout this process, they leveraged both collaborative problem-solving and AI-assisted tools to verify answers, optimize outcomes, and enhance their final products, demonstrating integrated application of cognitive, social, and technological resources in their learning journey. Table 2 manifests examples of CTACLL activities performed in class.

Study design

In this study, the researchers employed a bisymmetric research design, which was introduced to the fields of education and CALL by Rahimi and Sevilla-Pavón (2024; 2025a). In this context, the symmetrical phase of the study adopts a mean-based quantitative approach grounded in sufficiency logic to determine should have factors that contribute to the desired outcomes. Conversely, the asymmetrical part follows the necessary logic, examining must-have factors and necessary conditions that must be present in the study to achieve the intended results (Rahimi & Sevilla-Pavón, 2024), as shown in the Fig. 1. Accordingly, PLS-SEM was used as a part of the symmetrical phase of the study to explore direct and serial correlations between the study variables. We applied it because it can evaluate complex relationships with small sample sizes due to its robustness against violations of normality assumptions (Sarstedt et al., 2020). Furthermore, PLS-SEM can be used to assess the effects of moderation and mediation simultaneously (Hair et al., 2021). Additionally, it facilitates the implementation of higher-order constructs, allowing simultaneous specification of constructs at several levels of abstraction (Hair et al., 2019). In contrast, NCA was implemented as an asymmetrical part to explore the necessary conditions that must be present to achieve the target outcome.

Results

The symmetrical Phase

Because the data for this study were collected from a single source (self-report questionnaires) at a single time point (Podsakoff et al., 2003), there is a possibility of common method bias (CMB). To address this problem, we conducted Harman's single-factor test, which resulted in a score of 23.61, below the recommended threshold of 50% (Podsakoff & Organ, 1986). Then, in the reflective model of PLS-SEM, we assessed the factors' loading, reliability, and convergent validity. In terms of reliability, Cronbach's alpha and composite reliability were evaluated, with Hair et al. (2021) suggesting a threshold of over 0.70 for both. For convergent validity, we applied the average variance extracted (AVE), which should be greater than 0.50. As shown in Table 3, all latent variables were valid and reliable.

Discriminant validity was evaluated using the Heterotrait-Monotrait ratio (HTMT) introduced by Henseler et al. (2015), which determines whether a construct is empirically distinct from others in the model. Henseler et al. (2015) suggest that the average HTMT value should not exceed 0.90. The findings presented in Table 4 confirm this criterion.

To find the correlations between CTCALL and language learners' motivation, the bootstrap was used, which showed that AB significantly predicts AUT and CU.

Table 2 Examples of activities, AI (assisted) tools, and CTCALL skills for each text

Stage	Activities	AI (assisted) (corpus) tools	CT skills
Engagement	Guessing the general idea of the text	VERSATILE	<i>Abstraction</i> Working out the topic of the text based on the AI-generated vocabulary clouds based on the frequency of the words, and reading the relevant sentences instead of reading the whole text in detail <i>Algorithmic Thinking</i> Step-by-step processes each of which paving the way for the next one in working out the gist of the text (In the case of identifying the genres of texts, first they were exposed to multimodal texts introducing genres and related AI-generated pictures exemplifying each genre, then they looked at the whole sentences for the most frequent words in AI-generated word clouds and decided on the genre) <i>Generalization</i> Employment of their learnt vocabulary and background knowledge to discuss the topic <i>Collaboration</i> Collaboration with classmates and AI tools to generate topic-related words or generate vocabulary for AI-generated media
	Identifying the genre of the text		
	Brainstorming the new relevant vocabulary	*Steve.ai, PIXLR, Canva, Fotor, and other similar tools ChatGPT/ DeepSeek/ Grok/ Google search engine	

Table 2 (continued)

Stage	Activities	AI (assisted) (corpus) tools	CT skills
Study	Responding to different types of reading comprehension check questions	*Murf and similar tools to change the target texts to audio *Twee and similar tools ChatGPT/ DeepSeek	<i>Algorithmic Thinking</i> Step-by-step employment of strategies in identifying the answers in the text <i>Generalization</i> Application of the previously learnt concepts in identifying the key vocabulary in the stem, identifying either the target word in the text, paying attention to the negative words, adjectives, and adverbs, and concluding words to identify and report on the final joint responses <i>Collaboration</i> Cooperation in the group while referring to different AI tools, and finally collaborating with the AI-generated audio for reporting on the joint response with accurate pronunciation
			<i>Algorithmic Thinking</i> Step-by-step analysis of the vocabulary's definition in dictionaries, their examples, collocates, synonyms, or similar words through word clouds using AI tools, and finally, consideration of the context in the AI-generated stems by Twee and similar AI-assisted websites <i>Generalization</i> Recognizing the patterns in examples and collocations <i>Collaboration</i> Cooperation in their groups with the members and different tools to reach the best joint answer <i>Evaluation</i> Evaluating the information received from different tools, as well as consulting more proficient members, or even checking the fit of the response by AI tools
	Paraphrasing the challenging sentences	ChatGPT/ DeepSeek/ Grok/ Google Translate/ Online dictionaries	<i>Collaboration</i> Cooperatively collaborating in their groups with different tools and exchanging ideas to find the best paraphrase <i>Evaluation</i> Critically evaluating the answers provided by AI tools, the more competent peers, Google Translate, and online dictionaries to reach the best response

Table 2 (continued)

Stage	Activities	AI (assisted) (corpus) tools	CT skills
Activation	Presenting a final story for the AI-generated pictures to activate the new vocabulary, structures, and concepts	*Steve.ai, PIXLR, Canva, Fotor, and other similar tools Chat-GPT/ DeepSeek/ Grok/ Google Translate	<i>Abstraction</i> Summarizing insights and ideas gained from the texts and trying to simplify them while relating them to the pictures <i>Generalization</i> Employing the learnt vocabulary, concepts, and structures
	Producing written summaries of the texts	Grammarly, ChatGPT/ DeepSeek	<i>Algorithmic Thinking</i> Step-by-step following the guidelines presented through a watched video in class <i>Abstraction</i> Summarizing the insights and recognizing the major points in the text <i>Evaluation</i> Improvement of the final product based on the feedback received from the instructor and peers in class, Grammarly, ChatGPT, or DeepSeek

*The specified points with a star have been used to show the tools employed for generating materials by the instructor *Twee has been occasionally used, depending on whether the AI produced the questions or the instructor designed them

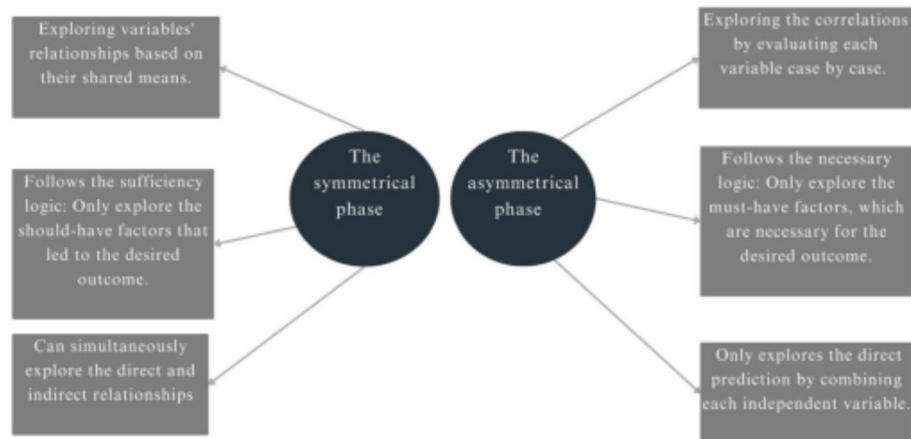


Fig. 1 The differences between symmetrical and asymmetrical phases (Rahimi & Sevilla-Pavón, 2026)

Table 3 The validity and reliability results

Variable	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
AB	0.834	0.839	0.900	0.750
AL	0.859	0.940	0.901	0.754
AUT	0.840	0.840	0.903	0.757
COL	0.816	0.928	0.885	0.720
CU	0.751	0.752	0.859	0.672
DA	0.974	0.975	0.987	0.974
EV	0.860	0.872	0.915	0.782
GE	0.708	0.809	0.839	0.647
IDL	0.790	0.794	0.877	0.705
OR	0.707	0.717	0.872	0.772
OUT	0.743	0.745	0.855	0.665
SA	0.769	0.863	0.892	0.806

Table 4 HTMT

	AB	AL	AUT	COL	CU	DA	EV	GE	IDL	OR	OUT	SA
AB												
AL	0.381											
AUT	0.408	0.156										
COL	0.104	0.097	0.091									
CU	0.508	0.378	0.630	0.149								
DA	0.537	0.389	0.590	0.067	0.682							
EV	0.114	0.140	0.512	0.099	0.582	0.411						
GE	0.549	0.338	0.597	0.058	0.806	0.659	0.458					
IDL	0.285	0.160	0.498	0.097	0.750	0.340	0.311	0.531				
OR	0.087	0.131	0.077	0.072	0.234	0.065	0.085	0.141	0.362			
OUT	0.197	0.202	0.491	0.110	0.631	0.463	0.645	0.550	0.176	0.071		
SA	0.047	0.283	0.197	0.095	0.060	0.071	0.111	0.168	0.138	0.049	0.108	

Moreover, the AL could only predict CU, while the EV significantly predicted language learners' CU. Additionally, GE was the only CTCALL skill that positively correlated with all the AIL2MS factors. Similarly, EV positively correlated with all motivational factors except IDL. The results of the structural correlation between CTCALL and AIL2MS are presented in Table 5.

To address the study's third research question, a direct analysis with 5000 subsamples was conducted to explore the direct effect of CTCALL on language learners' approaches to ICALL. As shown in Table 6, only abstraction ($\beta=0.139$), evaluation ($\beta=0.254$), and generalization ($\beta=0.230$) were found to directly predict language learners' deep approach to ICALL.

Furthermore, the bootstrap indirect analysis as shown in Table 7 revealed that only the digital self-authenticity and current L2-self ($GE > CU > DA$; $\beta=0.164$); ($EV > CU > DA$;

Table 5 The correlation between CTCALL and AIL2MSS

Paths	Beta	VIF	T statistics	P values
AB—> AUT	0.224	1.266	2.421	0.016
AB—> CU	0.171	1.266	2.261	0.024
AB—> IDL	0.096	1.266	0.958	0.338
AB—> OUT	0.004	1.266	0.037	0.971
AL—> AUT	−0.029	1.142	0.300	0.764
AL—> CU	0.178	1.142	2.266	0.023
AL—> IDL	0.007	1.142	0.068	0.946
AL—> OUT	0.088	1.142	0.956	0.339
AUT—> DA	0.323	1.459	2.494	0.013
AUT—> OR	−0.198	1.459	2.041	0.041
AUT—> SA	0.235	1.459	1.930	0.054
COL—> AUT	−0.019	1.019	0.190	0.849
COL—> CU	−0.067	1.019	0.758	0.448
COL—> IDL	−0.070	1.019	0.688	0.492
COL—> OUT	0.010	1.019	0.100	0.920
CU—> DA	0.444	2.107	3.543	0.000
CU—> OR	0.144	2.107	1.165	0.244
CU—> SA	−0.083	2.107	0.541	0.588
EV—> AUT	0.328	1.192	3.234	0.001
EV—> CU	0.295	1.192	3.797	0.000
EV—> IDL	0.131	1.192	1.495	0.135
EV—> OUT	0.418	1.192	4.686	0.000
GE—> AUT	0.249	1.447	2.140	0.032
GE—> CU	0.370	1.447	4.278	0.000
GE—> IDL	0.316	1.447	2.496	0.013
GE—> OUT	0.225	1.447	2.265	0.024
IDL—> DA	−0.096	1.667	0.908	0.364
IDL—> OR	0.277	1.667	2.173	0.030
IDL—> SA	0.075	1.667	0.567	0.570
OUT—> DA	0.070	1.442	0.766	0.444
OUT—> OR	−0.071	1.442	0.599	0.549
OUT—> SA	−0.138	1.442	0.997	0.319

Table 6 The direct analysis

Paths	Beta	2.5%	97.5%	P-value
AB—> DA	0.139	0.047	0.253	0.008
AB—> OR	0.007	−0.080	0.102	0.883
AB—> SA	0.045	−0.042	0.134	0.306
AL—> DA	0.075	−0.031	0.194	0.186
AL—> OR	0.027	−0.048	0.113	0.500
AL—> SA	−0.033	−0.138	0.032	0.440
COL—> DA	−0.028	−0.136	0.092	0.629
COL—> OR	−0.026	−0.107	0.058	0.530
COL—> SA	−0.006	−0.078	0.067	0.878
EV—> DA	0.254	0.125	0.374	0.000
EV—> OR	−0.016	−0.124	0.082	0.763
EV—> SA	0.004	−0.115	0.126	0.943
GE—> DA	0.230	0.117	0.355	0.000
GE—> OR	0.076	−0.046	0.199	0.214
GE—> SA	0.020	−0.084	0.143	0.728

$\beta=0.131$) mediated the relationship between learners' CTCALL and their approaches to AILL, and the rest couldn't work as a mediator variable.

To assess the model's predictive power, the coefficient of determination was used to estimate the contribution of exogenous variables and mediator variables to endogenous variables. Using Hair et al. (2023) classification, predictive power is categorized as weak, moderate, or strong at values of 0.19, 0.37, and 0.67, respectively, all of which were achieved at sufficient levels in our study model. Furthermore, Stone-Geisser's (Q^2) was calculated to evaluate predictive accuracy, with an acceptable relevance level defined as a value above 0, which was met in the study model. As shown in Table 8, the SRMR was below 0.08, indicating a good model fit (Hair et al., 2020). In Fig. 2, the study model is illustrated with the corresponding relative values.

The asymmetrical phase

Unlike symmetrical approaches, asymmetrical approaches depend on normality. Thus, we evaluated the normal distribution of latent variables' scores, which were distributed normally between −1 and 1, as shown in Table 9.

Then NCA permutation was conducted, utilizing a ceiling-envelopment-free disposal hull (CE-FDH) due to its non-decreasing stepwise linear function, which is more precise than ceiling regression. As shown in Table 4, none of the variables had a significant impact on language learners' organizing approach to ICALL based on the NCA permutation. However, regarding the deep approach, the learners' abstraction and algorithmic thinking skills were significant predictors of their deep approach to, with effect sizes of 0.1 and 0.21, indicating small and medium effects, respectively (Dul, 2023), as shown in Tables 10 and 11.

Based on the asymmetrical and symmetrical results, we achieved the following scenarios: The first scenario was that for the organizing approach, none of the CTCALL and AILL2MS factors were among the necessary conditions to shape language learners' organizing approach, while the presence of IDL, OUT, and AUT can escalate this process. The

Table 7 Result of the mediation analysis

Paths	Beta	2.5%	97.5%	T statistics (O/ STDEV)	P values
AB→OUT→DA	0.000	−0.022	0.025	0.022	0.982
AL→IDL→DA	−0.001	−0.030	0.033	0.045	0.964
AB→OUT→OR	0.000	−0.027	0.030	0.019	0.985
COL→IDL→DA	0.007	−0.017	0.053	0.397	0.691
AB→OUT→SA	0.000	−0.041	0.042	0.025	0.980
EV→OUT→DA	0.029	−0.044	0.107	0.769	0.442
AL→CU→DA	0.079	0.008	0.182	1.750	0.080
AL→IDL→OR	0.002	−0.053	0.072	0.061	0.951
AB→AUT→DA	0.072	0.005	0.177	1.612	0.107
AL→IDL→SA	0.001	−0.040	0.029	0.032	0.975
COL→CU→DA	−0.030	−0.101	0.049	0.781	0.435
COL→IDL→OR	−0.019	−0.091	0.044	0.595	0.552
GE→IDL→DA	−0.030	−0.111	0.039	0.814	0.415
EV→OUT→OR	−0.030	−0.138	0.062	0.583	0.560
AL→CU→OR	0.026	−0.017	0.092	0.920	0.358
COL→IDL→SA	−0.005	−0.058	0.024	0.269	0.788
AB→AUT→OR	−0.044	−0.121	0.002	1.401	0.161
EV→OUT→SA	−0.058	−0.172	0.061	0.982	0.326
AL→CU→SA	−0.015	−0.096	0.036	0.453	0.650
COL→CU→OR	−0.010	−0.054	0.021	0.514	0.607
GE→CU→DA	0.164	0.052	0.290	2.694	0.007
GE→IDL→OR	0.088	−0.004	0.200	1.680	0.093
AB→AUT→SA	0.053	−0.008	0.125	1.540	0.124
EV→AUT→DA	0.106	0.022	0.199	2.342	0.019
COL→CU→SA	0.006	−0.025	0.052	0.300	0.764
GE→IDL→SA	0.024	−0.053	0.143	0.486	0.627
GE→CU→OR	0.053	−0.040	0.157	1.084	0.279
EV→AUT→OR	−0.065	−0.154	−0.001	1.654	0.098
GE→CU→SA	−0.031	−0.151	0.088	0.526	0.599
EV→AUT→SA	0.077	−0.005	0.186	1.574	0.116
AL→OUT→DA	0.006	0.012	0.047	0.419	0.675
AB→IDL→DA	−0.009	−0.049	0.023	0.532	0.595
COL→OUT→DA	0.001	−0.026	0.024	0.060	0.952
AL→OUT→OR	−0.006	−0.048	0.023	0.351	0.726
AB→CU→DA	0.076	0.006	0.174	1.740	0.082
AB→IDL→OR	0.027	−0.025	0.110	0.789	0.430
AL→OUT→SA	−0.012	−0.077	0.017	0.503	0.615
COL→OUT→OR	−0.001	−0.034	0.029	0.050	0.960
GE→OUT→DA	0.016	−0.023	0.073	0.648	0.517
AB→IDL→SA	0.007	−0.032	0.053	0.365	0.715
EV→IDL→DA	−0.013	−0.051	0.020	0.714	0.475
AL→AUT→DA	−0.009	−0.081	0.058	0.279	0.780
COL→OUT→SA	−0.001	−0.047	0.036	0.071	0.944
AB→CU→OR	0.025	−0.018	0.082	0.972	0.331
COL→AUT→DA	−0.006	−0.070	0.065	0.187	0.852
GE→OUT→OR	−0.016	−0.084	0.037	0.531	0.595
AB→CU→SA	−0.014	−0.073	0.046	0.498	0.619
EV→CU→DA	0.131	0.041	0.240	2.547	0.011
EV→IDL→OR	0.036	−0.015	0.114	1.099	0.272

Table 7 (continued)

Paths	Beta	2.5%	97.5%	T statistics (O/STDEV)	P values
AL→AUT→OR	0.006	−0.039	0.054	0.263	0.793
GE→OUT→SA	−0.031	−0.121	0.030	0.829	0.407
EV→IDL→SA	0.010	−0.029	0.060	0.454	0.650
AL→AUT→SA	−0.007	−0.067	0.041	0.263	0.792
COL→AUT→OR	0.004	−0.040	0.052	0.169	0.866
GE→AUT→DA	0.080	0.001	0.224	1.363	0.173
EV→CU→OR	0.043	−0.033	0.120	1.111	0.267
COL→AUT→SA	−0.004	−0.057	0.053	0.173	0.863
EV→CU→SA	−0.025	−0.123	0.075	0.513	0.608
GE→AUT→OR	−0.049	−0.132	0.005	1.380	0.168
GE→AUT→SA	0.058	−0.007	0.172	1.235	0.217

Table 8 Model fit

Q^2	R^2	Variables
0.223	0.331	AUT
0.308	0.502	CU
0.400	0.430	DA
0.104	0.193	IDL
0.03	0.112	OR
0.18	0.327	OUT
0.01	0.058	SA

SRMR=0.06

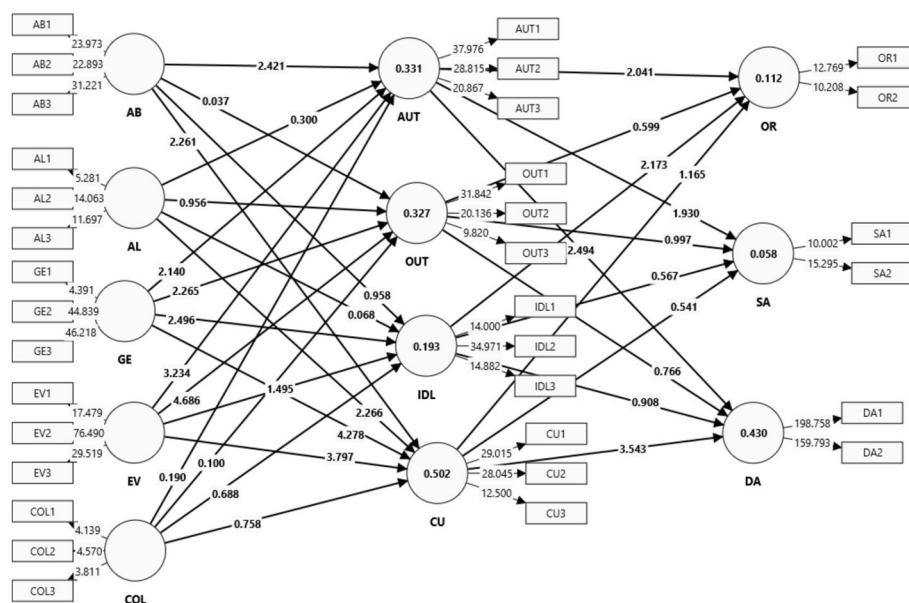

Fig. 2 The structural model with relative values

Table 9 Latent variables' scores' normality distribution

Latent variables' scores	Mean	Median	Observed min	Observed max	Excess kurtosis	Skewness
AL	0.000	− 0.077	− 2.108	1.889	− 0.683	− 0.021
OUT	0.000	0.032	− 2.238	2.403	− 0.402	− 0.114
CU	0.000	0.223	− 2.200	2.109	− 0.527	− 0.254
EV	0.000	− 0.037	− 1.904	2.460	− 0.294	− 0.263
AB	− 0.000	0.129	− 2.734	2.064	− 0.206	− 0.239
GE	− 0.000	0.078	− 2.084	2.183	− 0.395	− 0.070
OR	0.000	− 0.230	− 2.142	2.550	0.231	0.511
COL	− 0.000	0.170	− 2.780	1.981	− 0.402	− 0.309
AUT	− 0.000	0.181	− 2.816	2.877	0.674	− 0.031
IDL	− 0.000	− 0.117	− 2.194	2.746	− 0.225	0.129
DA	− 0.000	0.036	− 2.215	2.286	− 0.599	0.061

Table 10 The NCA permutation for the organizing approach

Latent variables' scorers	Original effect size	95.0%	Permutation <i>p</i> -value
LV scores—AB	0.198	0.215	0.100
LV scores—AL	0.061	0.163	0.674
LV scores—AUT	0.136	0.198	0.408
LV scores—COL	0.183	0.224	0.233
LV scores—CU	0.111	0.154	0.251
LV scores—EV	0.072	0.117	0.229
LV scores—GE	0.084	0.141	0.345
LV scores—IDL	0.135	0.142	0.076
LV scores—OUT	0.012	0.149	0.903

Table 11 The NCA permutation for the deep approach

	Original effect size	95.0%	Permutation <i>p</i> -value
LV scores—AB	0.254	0.211	0.003
LV scores—AL	0.169	0.164	0.040
LV scores—AUT	0.178	0.194	0.084
LV scores—COL	0.206	0.226	0.111
LV scores—CU	0.133	0.163	0.154
LV scores—EV	0.072	0.118	0.316
LV scores—GE	0.065	0.143	0.611
LV scores—IDL	0.088	0.142	0.437
LV scores—OUT	0.128	0.150	0.134

second scenario is that algorithm thinking skill was among the necessary conditions to shape learners' DA; however, a further level of it cannot manifest this learning behavior. In the third scenario, we found that the abstraction skill was among the necessary conditions in shaping language learners' DA to ICALL, and a further level of it can manifest this process as well. In other words, at lower desired achievement levels (0–40%), very

few factors act as bottlenecks, with only AB consistently necessary and AL becoming relevant above 20%.

Furthermore, evaluation and generalization skills were not among the necessary conditions to shape learners' DA; however, a further level of them can assist this process. Moreover, the IDL and OUT were neither among the necessary factors nor should have been factors that can shape language learners' DA to ICALL. And as a last one, the CU and AUT were not among the necessary conditions to shape learners' DA; however, a further level of it can manifest this learning behavior. Table 13 summarizes the following scenarios.

The necessary condition analysis reveals that achieving higher levels of the deep approach requires meeting minimum thresholds across an increasing number of conditions. At lower desired achievement levels (0–40%), very few factors act as bottlenecks, with only AB consistently necessary and AL becoming relevant above 20%. There is, however, a significant shift occurring at the 50% threshold, where multiple conditions, including COL, IDL, and OUT, become necessary to achieve moderate success. Increasing the desired outcome to 70% or higher creates additional bottlenecks, including AUT, CU, and EV, while previously essential factors like AB and AL must be maintained at specific levels. In the case of the highest levels of achievement (90–100%), nearly all conditions are necessary, but AUT requires a particularly high level of achievement (1.444), whereas at lower levels it is irrelevant. Table 12 displays the necessary conditions for the deep approach.

Based on the asymmetrical and symmetrical results, we achieve the following scenarios:

The first scenario was that for the organizing approach, none of the CTCALL and AIL2MS factors were among the necessary conditions to shape language learners' organizing approach, while the presence of IDL, OUT, and AUT could escalate this process. The second scenario is that algorithm thinking skill was among the necessary conditions to shape learners' DA, and a further level of it can manifest this learning behaviour. In the third scenario, we found that the abstraction skill was among the necessary conditions in shaping language learners' DA to ICALL, and a further level of it can manifest this process as well.

Furthermore, evaluation and current L2-self were not the necessary conditions to shape learners' DA, unless they want to achieve 70 percent of it. Moreover, IDL, OUT,

Table 12 Deep approach's bottleneck results

	DA	AB	AL	AUT	COL	CU	EV	GE	IDL	OUT
0.000%	−2.215	NN	NN	NN	NN	NN	NN	NN	NN	NN
10.000%	−1.765	−1.934	NN	NN	NN	NN	NN	NN	NN	NN
20.000%	−1.314	−1.706	NN	NN	NN	NN	NN	NN	NN	NN
30.000%	−0.864	−1.706	NN	NN	NN	NN	NN	NN	NN	NN
40.000%	−0.414	−1.706	−1.296	NN	NN	NN	NN	NN	NN	NN
50.000%	0.036	−1.250	−1.296	NN	−1.640	NN	NN	NN	−2.106	−1.115
60.000%	0.486	−1.250	−1.296	NN	−1.640	NN	NN	NN	−2.106	−1.115
70.000%	0.936	−1.250	−1.296	−1.133	−1.488	−0.617	−1.600	NN	−1.742	−1.017
80.000%	1.386	−1.250	−1.296	−1.133	−1.488	−0.617	−1.600	NN	−1.742	−1.017
90.000%	1.836	−1.250	−0.484	1.544	0.690	−0.331	−0.319	−0.420	−0.139	−1.017
100.000%	2.286	−1.250	−0.484	1.544	0.690	−0.331	−0.319	−0.420	−0.139	−1.017

NN stands for not necessary conditions

Table 13 Relevant scenarios for the deep approach

Variable	Symmetric phase	Asymmetric phase	Conclusion
AB	Significant	High necessary	A certain level of abstraction skills is necessary for developing learners' DA, and increasing this level further enhances it
AL	Not significant	High Necessary	To foster learners' deep approach, a specific level of algorithmic thinking activities must be incorporated into the language class
EV	Significant	Not necessary	A certain level of evaluation skill is not necessary for developing learners' DA; however, a further level of it can shape their DA
GE	Significant	Not necessary	A certain level of generalization skill is not necessary for developing learners' DA; however, a further level of it can shape their DA
COL	Not significant	Not necessary	A certain degree of collaboration is neither necessary nor can it assist in shaping learners' DA
IDL	Not significant	Not necessary	A certain degree of learners' IDL is neither necessary nor can it assist in shaping learners' DA
CU	Significant	Not necessary	A certain degree of learners' CU is not necessary to shape their DA; however, its presence can manifest this process
AUT	Significant	Not necessary	A certain degree of learners' AUT is not necessary to shape their DA; however, its presence is necessary at the highest level (70 percent)
OUT	Not significant	Not necessary	To achieve a deep approach to ICALL, learners' OUT does not need to be at a certain level; however, to reach 50% of the deep approach, learners' OUT must be achieved (covered)

and COL were among the moderately necessary variables, meaning that they were necessary if we wanted to reach 50 percent of the deep approach. COL and OUT were not directly shaped learners' deep approach; however, they were among the moderate necessary conditions to shape learners' deep approach at 50 percent. And as the last one, CU and AUT were not among the necessary conditions to shape learners' DA; however, at a higher level, their presence is necessary. Table 13 summarizes the following scenarios.

Discussion

The factorial validity of CTCALL, and its correlation with learners' motivation, and their learning approaches to ICALL

Based on the results of the PLS-SEM's measurement model, the factorial structure of CTCALL has been validated in another area of CALL, specifically in ICALL, and the Iranian EFL context. This validation follows similar findings in VE and the Spanish EFL context (Rahimi & Sevilla Pavón, 2025), and we have now extended the factorial validity of this theory to the Iranian EFL context and ICALL. Furthermore, the results indicated that when learners used deductive reasoning to focus on key information provided by AI, rather than getting caught up in details while tackling language tasks, they perceived ICALL as significantly more authentic learning environments than previous ones. This has also contributed to their satisfaction with their current needs. It is important to note that the abstraction skill is the basic and most important (based on NCA result) CTCALL skill for shaping learners' deep approach to ICALL. In accordance with NCA, AB is required at every level of achievement, even the lowest, whereas skills such as evaluation and generalization are only necessary at intermediate levels. More importantly, it

is only abstraction that emerges as a necessary condition at the lowest levels of the deep approach (0–40%). This is consistent with a complex dynamic system's emphasis on initial conditions, which are the basis of the system's evolution (de Bot et al., 2007). A potential "seed" for deeper engagement with AI appears to be the ability to focus on key information without becoming distracted by details. In the absence of this basic cognitive orientation, the system cannot even begin to move toward a deep approach, regardless of other skills or motivations present. Abstraction may play a key role in providing a cognitive gateway into meaningful AI interaction, as students can focus on the core communicative goal, key themes, and structural requirements in order to avoid getting distracted by surface details at the beginning (Rahimi & SevillaPavón, 2025a). With this high-level framing, AI prompts will be of higher quality, resulting in more useful AI responses.

This might be a response to the problem raised by previous scholars who reported that AI can lead language learners to be passive and decrease their critical thinking skills (Mohebbi, 2024); however, teachers can mitigate it by providing several strategies, and CTCALL and abstraction skills can be one of them. It also had the strongest shared variance with learners' perceived authenticity gap, which may be due to AI's ability to provide several pieces of information from any data and learners' ability to analyze their accuracy based on abstraction skills and choose their required information among them based on their specific needs, which may not be feasible in traditional teacher-led lessons due to limited resources, time, and access to relevant data based on the target context, as previously reported in CALL studies (Assassi, 2025; Zou et al., 2023). This aligns with the claims of the pioneers of CT, who asserted that CT can effectively foster cognitive development if students actively engage in building knowledge through computers that give concrete form to intangible and abstract concepts (Papert, 1980).

Another surprising finding is the differential role of algorithmic thinking. In fact, AL did not significantly predict the deep approach in the symmetrical PLS-SEM analysis (see Table 6), but in the asymmetrical NCA, it was revealed as a high-level necessary condition. This apparent contradiction is a hallmark of bisymmetric research and underscores the distinction between causal sufficiency and causal necessity (Rahimi & Sevilla-Pavón, 2024, 2026). In light of the non-significant path, it seems likely that being skilled at solving language tasks using AI in a step-by-step manner does not guarantee that the learner will engage in the deep, cross-referencing thought that characterizes deep learning. Possibly, this is due to AL's high time demands and cognitive demands (Adorni et al., 2024); learners may possess it but not use it due to faster, heuristic-based interactions with AI. Although algorithmic thinking is a necessary condition, its status as one of the necessary conditions suggests a more fundamental truth: a deep approach to ICALL is impossible without algorithmic thinking. According to NCA's bottleneck analysis (Table 11), achieving even a moderate level (40–50%) of deep approach requires a certain level of AL. The absence of this threshold prevents learners from engaging deeply and meaningfully with AI. Accordingly, algorithmic thinking functions as a control parameter for certain system states, but it does not dictate the system's behaviour once the threshold has been crossed (Hiver & Al-Horie, 2019). Based on a complex dynamic system, AL represents a necessary but not sufficient initial condition. The system must reach even moderate levels of deep approach, but once that threshold has been reached, other interconnected factors (EV, GE, AUT, CU) will become the main

drivers of development. This distinction between enabling conditions and driving forces is precisely what a complex dynamic system predicts for complex, multi-causal systems.

Similarly, algorithmic thinking skill failed to predict language learners' L2MSSAILL, except for current L2-self. Consequently, the more language learners engage with AI to solve language tasks in a stepwise manner, the greater their satisfaction with their present language learning needs. This result is likely explained by the fact that AL is a highly time-consuming computational thinking skill with a high cognitive load, and that learners are increasingly favoring fast and optimal solutions rather than slow ones based on practicality, efficient cognitive shortcuts, or heuristics in learning with AI (Zhai et al., 2024). Carroll et al. (2015) demonstrated that individuals with a good understanding of core current competencies can systematically use upwardly revised possible selves in any language setting, and algorithmic thinking skill, as one of CTCALL's systematic skills, may assist language learners in revising their current self in ICALL.

Moreover, the evaluation skill had a positive correlation with language learners' AIL2MSS, including their AUT, CU, and OUT, and significantly predicted their deep approach to ICALL. Thereby, the higher language learners' have skills to compare various solutions and recognize the most appropriate response provided by AI; the more they can fulfill their expectations and self-assess their present learning achievements in ICALL, which led them to stimulate new ways of thinking and cross-reference the ideas with the assistance of AI (deep approach), inferring it as an authentic language learning environment compared to previous language learning contexts, culminating in maximizing its use for language learning. This finding is in line with Parsazadeh et al. (2020), who found that CT can increase language learners' external motivation in digital storytelling. Additionally, the findings are not at variance with those of Rahimi and Sevilla Pavón (2025), who found that evaluation skills did not contribute to Spanish learners' attitudes in general and to their environmental and contextual fitness in VE in particular. There may be a difference in the way students learn languages at VE and AI, as the former offers a collaborative learning experience, while the latter offers a personalized language learning experience where students are exposed to a vast amount of information provided by AI, among which they must make their own choices. However, they may experience inaccuracies, hallucinations, and potential bias while using AI (Crompton & Burke, 2024), and through CT's evaluation skill, learners may be able to compare information with each other's and their reliability, leading them to cross-refer ideas, compare them, and select the most appropriate one, and maximize the use of AI for language learning. Possibly, this evaluation skill is a response to AI bugs that provide fake information and references in language learning (Crompton & Burke, 2024).

A positive correlation with all AIL2MSS components with GE indicates that generalization is a critical condition for fostering language learners' deep learning approach within ICALL contexts. The presence of this generalization can manifest in enhanced learning behaviors. Specifically, when learners utilize AI to identify patterns in targeted language tasks and apply these patterns across diverse tasks and contexts, they are more likely to attain their aspirational goals. Furthermore, this CTCALL subskill supports language learners in meeting academic and personal objectives while enhancing the authenticity of ICALL experiences compared to earlier methods. By recognizing and transferring these patterns, learners can enhance AI use in language acquisition,

fostering innovative and critical thinking skills while enabling effective cross-referencing of ideas to maximize their AI use for learning language. These findings are in line with and extend a previous CALL study that reported generalization in CTCALL is the most powerful skill in shaping language learners' fitness with VE and their attitudes toward it (Rahimi & SevillaPavón, 2025a), and now we validate and extend their findings to ICALL, AIL2MS, and learners' language learning approaches. This is also converging with the interest-to-effort ratio (Smarandache et al., 2021): A deeper approach can be achieved by spending more time and effort on interesting materials and tasks, whereas a surface approach can be achieved by learning uninteresting materials with little effort. Several previous CALL studies have indicated that language learners prefer to learn languages with AI (Creely et al., 2025; Kartal & Yeşilyurt, 2024; Koç & Savaş, 2024), particularly when more patterns and skills can be generated through it.

All in all, CTCALL factors appear to correlate more strongly with deep approaches than organizing approaches, which might be related to the fact that deep approaches have the characteristics of intent to understand, critical engagement with ideas, meaningful learning, and integration of acquired knowledge with existing knowledge. (Entwistle, ; Marton & Säljö, 1976) CTCALL skills, like particularly abstraction, evaluation, and generalization, do not merely represent technical abilities but are cognitive operations that enact precisely these deep processes. As a matter of fact, when learners practice abstraction, they actively select and prioritize information—the key to gaining a deeper understanding of the material. It is during the evaluation of multiple AI responses that they engage in critical comparison and judgment, which is at the core of deep learning. In contrast, the organizing approach focuses on managing time, effort, and study environments (Entwistle, 2017). Despite their value, these regulatory activities are conceptually distinct from the cognitive actions that CTCALL skills entail. In organizing approaches, learners are concerned with the way they structure their learning; in CTCALL skills, learners are concerned with the way they process information with ICT.

The mediation role of AI L2 motivational self-system on the relationship between language learners' computational thinking skills in CALL and their approaches to ICALL

The bootstrap indirect analysis revealed that language learners' current L2-self and digital self-authenticity served as complementary mediators (GE->CU->DA; EV->AUT->DA; EV->CU->DA); however, ideal L2 self and ought-to L2 self could not act as mediators (see Table 7). Our interpretation of this finding is based on the concept of temporal alignment within complex dynamic systems theory (Salet et al., 2020). CTCALL skills, particularly evaluation and generalization, are exercised in the immediate present of interacting with AI. Learning through these approaches alters the current experiential quality of learning by enabling learners to judge AI outputs critically (EV) and transfer patterns across contexts (GE). In other words, these are proximal processes that have a direct impact on how learning feels now (Larsen-Freeman, 2017).

This concept of digital self-authenticity is particularly relevant to contemporary experiential learning, or in other words, comparing learners' present experiences with the current learning environment and the previous learning environment. By strengthening learners' agency, self-reliance, and critical engagement, CTCALL skills simultaneously amplify this sense of authenticity—a direct, immediate connection between learners and their learning

environment. Alternatively, the ideal L2 self and the ought-to L2 self are distal constructs, where the first one represents a future aspirational state and the second one reflects external obligations. This future-oriented self may influence long-term persistence or initial engagement decisions; however, it is too distant from the moment-to-moment for CTCALL skills to be activated by it. From a CDST perspective, IDL and OUT function as dormant attractors, which do not influence behaviour unless they are triggered by a specific stimulus (Hiver & Al-Hoorie, 2020). Thus, when learners acquire new digital skills like CTCALL, the most proximal motivational outcome is an enhanced sense of authenticity in the present—a feeling that "this way of learning is more real than what came before"—and their current L2-self (their present language competence). The authenticity of the experience, in turn, fuels a deeper level of engagement. Future-oriented selves may play a greater role in sustaining motivation over time or in influencing enrolment decisions; however, they cannot facilitate the transfer of new skills into immediate engagement.

Recent CALL studies indicate that language learners aim to maintain their existing language knowledge and keep their digital skills up to date with global advancements (Creely et al., 2025; Dornburg & Davin, 2024). Thorsen et al. (2020) noted that learners strive for improvement by assessing their present language skills within their specific contexts. Furthermore, the "experiences of frustrated authenticity" (Henry & Cliffordson, 2017) lead to learners tending to invest more time and effort in digital contexts than in traditional ones, which may be linked to AI's anthropomorphism, personalization, and intelligence, which can evoke human-like qualities due to its capacity to provide timely, objective feedback without emotional biases. Coupled with CTCALL's generalization and evaluation skills, learners can not only update their present language skills in a digital learning environment but also acquire the heart of 21st-century digital skills, which can enhance their deep language learning approach to ICALL. These results agree with and extend previous CALL studies that underlined the mediation roles of current L2-self and digital self-authenticity in shaping language learners' VE self-regulation (Rahimi & Sevilla Pavón, 2025b) and AI L2 grits (Rahimi & Sevilla Pavón, 2025 c) in their learning approaches.

All in all, based on our results, CTCALL factors have much more correlations with the deep approach than organizing one, which might be related to.

Pedagogical and practical implications

The study's findings have several educational and practical implications for humanistic studies, applied linguistics, CALL, psycholinguistics, and ICALL. Firstly, the researchers adhered to the guidelines set by organizations such as ATC21S, OECD, the EU, CSTA, ISTE, and IEA by incorporating essential 21st-century digital skills into the CALL domain. They also followed earlier advice from CALL scholars to shift the focus from merely teaching language skills with CALL materials to cultivating 21st-century digital skills as well (Rahimi, 2023, 2024, 2025). By validating the factorial structures of CTCALL in another language learning context and the CALL field, the researchers expanded the validity of this new theoretical framework and addressed STEM researchers by demonstrating that CT is not exclusive to their field but can be applied across various domains, especially in CALL.

Recent studies have generally reported the positive role of CT on language learners' performance, outcome, or motivation (Parsazade et al., 2020; Wu et al., 2024; Yu et al., 2024); typically, they treated CT as a unitary construct and evaluated its effects from an overall

perspective. Using a bisymmetric design, the present study disentangles the differential roles of specific CTCALL subskills. Our findings reveal that not all CTCALL skills function equivalently; in fact, abstraction and evaluation emerge as necessary conditions for a deep approach, while algorithmic thinking is necessary but not sufficient, and generalization is sufficient but not necessary. Previous studies have noted that CT may have an impact on motivation (Parsazade et al., 2020; Wu et al., 2024), but no specific mechanism has been identified. In their studies, Parsazadeh et al. (2020) found that CT improved motivation in digital storytelling, but they did not identify which motivational constructs were affected or how they were affected. They only covered how CT can influence language learners' intrinsic and extrinsic motivation, but here we added current L2-self and digital self-authenticity as well. Additionally, by demonstrating the serial correlations among CTCALL, AILM2SS, and SAL, the researchers contributed a new insight to the complex dynamic system (Mercer, 2018) and furthered the development of the CALL field, as advocated by its pioneers (Coelpart, 2020). The importance of this integration lies in its response to the calls for research that captures the complex, multidimensional nature of learners' engagement with ICALL (Kartal & Yeşilyurt, 2024; Koç & Savaş, 2024).

Furthermore, by examining the relationship between language learners' CTCALL and their motivation and SAL, the study addressed previous gaps and redirected the focus from the link between CTCALL and learners' programming or computing skills to their learning approaches (Tang et al., 2020; Weng et al., 2024). Similarly, it shifted the emphasis from investigating language learners' attitudes and experiences, as well as their emotional factors related to AILL, to their pedagogical and cognitive aspects, which have been recommended by previous CALL studies (Kartal & Yeşilyurt, 2024; Koç & Savaş, 2024; Meina & Chaoran, 2025).

Moreover, our findings extend our understanding of L2MSS by showing that in AI-mediated contexts, motivational constructs play a mediating role according to their temporal orientation. While previous research in LMOOC (Rahimi, 2023) and emergency remote language learning (Rahimi & Mosalli, 2024) has found that learners' ideal and ought-to L2 selves predict their intentions in ICALL, these future-oriented selves remain dormant (Hiver, 2015). Digital self-authenticity and the current L2 self are two present-oriented constructs that actively influence the impact of CTCALL skills on the deep approach. Thus, AI-mediated learning activates different motivational subsystems than other digital contexts, which strengthens our understanding of how L2 selves operate across various learning environments and skills.

Our study reveals three key practical implications. Firstly, we advise language teachers to strike a balance between language skills and 21st-century digital skills, especially CTCALL. This is crucial in today's context, as AI increasingly plays a significant role in the daily and academic lives of language learners, who are now more inclined to seek solutions to their problems independently. Based on our findings, abstraction and algorithmic thinking skills are among the necessary conditions in shaping language learners' deep approach to ICALL; accordingly, to have language learners with effective ways of thinking and cross-referencing the language information with AI in a meaningful manner, language teachers must lead them to not blindly follow what AI provides to them but rather process their information in a stepwise manner and incidentally try to catch up on the most important and reliable information. Evaluation and generalization skills were

not essential for fostering a deep approach to ICALL; however, they were strong predictors of learners' motivation and deep learning in it. Therefore, we recommend enhancing both deep learning approaches and language motivation with these CTCALL skills as well. Thus, teachers should create opportunities for students to recognize patterns in addressing specific language tasks using AI and encourage them to apply these patterns to different language tasks and contexts. These CTCALL skills in AI significantly contribute to learners successfully completing their language courses, helping them envision an ideal future of language learning with AI, and viewing both CTCALL and AI as more authentic skills and contexts to learn language compared to previous ones.

Stakeholders, executive managers, and CALL experts should be aware of the digital self-authenticity that we found in the study context, where learners have much more motivation and a deep language learning approach to AI, particularly through CTCALL. As a result, they should incorporate 21st-century digital skills, such as CTCALL, into the language teaching procedure, curriculum, and books to prepare students for learning, living, and working in a globally advanced society that relies heavily on 21st-century digital skills. Additionally, language teachers should receive training on effectively integrating CTCALL skills into language education using CALL.

The study employed a bisymmetric research design, offering a novel pathway for CALL researchers to align with other disciplines that have integrated essential 21st-century digital skills. It also provides opportunities for replication, validation, and expansion of the existing conceptual framework within other CALL domains. Future research should address several limitations of the present study. As the data were collected exclusively through self-reported Likert-scale measures, future investigations could incorporate qualitative approaches, such as in-depth interviews or focus groups, to provide richer insights into the complex relationships among CTCALL, motivation, and SAL, and to examine both necessary and sufficient conditions underlying these associations.

Moreover, the relatively small and homogeneous sample ($N = 104$), consisting solely of female participants, limits the external validity of the results. Participants represented diverse academic majors and varying levels of language proficiency, which may introduce uncontrolled variability. The instructional context was also limited to a course primarily focused on reading comprehension; therefore, the applicability of the findings to other language skills (e.g., writing, listening, or speaking) remains to be established. Also, replicating the study across different countries may balance the localized characteristics reflected in the current study and add to the generalizability of the results. Furthermore, regional constraints affecting students' access to AI and AI-assisted tools may have influenced their engagement with CTCALL and related constructs. Future studies should therefore recruit larger, more gender-balanced, and more proficiency-controlled samples across varied instructional contexts, with more equitable technological access to strengthen the stability and representativeness of the findings.

Appendix

See Table 14.

Table 14 Study scale

Variable	Source	Item	Mean	Excess kurtosis	Skewness
AB1	Adapted from Rahimi and Sevilla-Pavón (2025)	When solving tasks with AI, I looked at the big picture instead of the details	4.798	−0.584	−0.354
AB2		It is more important to me to look at the most important aspects of solving language tasks with AI from a whole-point-of-view rather than focusing on details	4.058	−0.381	−0.005
AB3		When I want to solve language tasks with AI, I usually try to identify the key points	4.548	−0.545	−0.209
AL1		Whenever I had a task that I used AI for, I knew the steps I needed to follow	3.587	−0.530	0.026
AL2		I usually try to find effective solutions for solving language tasks with AI	3.596	−0.785	0.241
AL3		I usually try to lay out the steps for solving language tasks with AI	3.596	−0.695	−0.056
AUT1	Adapted from Rahimi (2023)	Learning English with artificial intelligence is more interesting than a face-to-face language class	4.000	0.178	−0.176
AUT2		Learning English with artificial intelligence motivates me more than a face-to-face course	4.135	0.221	−0.005
AUT3		The actual process of English language learning with artificial intelligence feels more real compared to the things we do in traditional language classes	4.288	−0.144	0.050
COL1	Adapted from Rahimi and Sevilla-Pavón (2025)	I like experiencing collaboration and solving tasks with others through AI	3.433	0.206	−0.307
COL2		I enjoy working cooperatively with my classmates on several language tasks through AI	3.471	−0.435	−0.069
COL3		I seek the optimal solution using AI when confronted with a difficult situation, like tackling a language task	3.538	0.697	0.254
CU1	Adapted from Rahimi (2023), and Smith et al. (2020)	Now, I see myself as someone who is good at learning English with artificial intelligence	3.721	−0.203	0.247
CU2		Being someone who can learn English with artificial intelligence is part of the person I am now	3.779	−0.356	0.298
CU3		Now, I see myself as someone who can use artificial intelligence to learn language in many different situations	3.615	−0.856	−0.372
DA1	Adapted from Rahimi (2024)	I try to use AI to develop my language proficiency	3.981	−0.573	0.081
DA2		I find using AI will help me to develop my language proficiency	3.923	−0.591	0.094

Table 14 (continued)

Variable	Source	Item	Mean	Excess kurtosis	Skewness
EV1	Adapted from Rahimi and Sevilla-Pavón (2025)	I usually think of the best answer for our tasks while using AI	3.865	−0.678	0.025
EV2		I usually think of a fast solution to solve the language tasks while using AI	4.067	−0.585	−0.266
EV3		I seek the optimal solution using AI when confronted with a difficult situation, like tackling a language task	4.192	−0.702	−0.126
GE1		My experience solving language tasks with AI can help me solve similar tasks in real life situations	3.740	−0.293	0.593
GE2		Based on my experience solving language tasks with AI, I can find the easiest way to solve different languages	4.327	−0.612	0.104
GE3		I can use well-known AI addressed solutions to tackle other language tasks	4.394	−0.374	0.231
IDL1	Adapted from Rahimi (2023), and Smith et al. (2020)	Learning English with artificial intelligence is essential for me as this language can make a difference in my life in the future	3.635	−0.356	0.080
IDL2		Learning English with Artificial Intelligence is essential for me because acquiring the language will change my life in the future	3.760	−0.794	0.118
IDL3		Learning English with artificial intelligence is important to me because an educated person is supposed to be able to speak English	4.125	−0.497	0.042
OR1	Adapted from Rahimi (2024)	consistently utilize AI as a language tool in my daily academic activities	3.731	−0.354	0.311
OR2		I regularly use AI as a language tool in my academic routine	3.837	−0.174	0.190
OUT1	Adapted from Rahimi (2023), and Smith et al. (2020)	I have to learn English with artificial intelligence so as to pass this course	4.058	−0.969	0.231
OUT2		I have to learn English with artificial intelligence to pass my final semester	4.048	−0.137	−0.164
OUT3		When thinking of not becoming a successful language learner with artificial intelligence, I feel scared	4.827	−0.354	−0.562
SA1	Adapted from Rahimi (2024)	I restrict my use of AI for learning the English language to as little as possible	2.904	−1.394	0.257
SA2		I did not find using AI to improve my language proficiency	2.692	−1.290	0.243

Author contributions

Author 1: Theoretical and methodological framework, study conception and design, analysis and interpretation of results, draft manuscript preparation, reviewing the results and approving the final version of the manuscript. Author 2: proofreading, investigation, study conception and design, reviewing the results and approving the final version of the manuscript.

Funding

A grant or other form of financial support was not received by the authors.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Declaration of generative AI in scientific writing

The authors in the current study did not use any kinds of artificial intelligence in any parts, and sections of the article.

Competing interests

Authors declare no conflicts of interest.

Received: 21 August 2025 Accepted: 10 March 2026

Published online: 08 April 2026

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